MIL-C-46535 (Ord)
12 May 1961

MILITARY SPECIFICATION
CASING, BURSTER, T21, ASSEMBLY

1. SCOPE

1.1 This specification covers Casing, Burster, T21, Assembly for Projectile, 81mm, Mortar, Chemical Loading.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military
- MIL-G-2550 - General Specification for Ammunition, Except Small Arms Ammunition
- MIL-I-45208 - Inspection Requirements, General Specification for
- MIL-P-10025 - Packing and Marking for Domestic Shipment of Inert Ammunition Components, General Specification for

STANDARDS

Federal
- Federal Test Method Standard No. 151 - Metals; Test Methods

Military
- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
- MIL-STD-109 - Inspection Terms and Definitions
- MIL-STD-639 - Surface Roughness Values Preferred for Army Ordnance Materiel

DRAWINGS

Ordnance Corps
- D-10520029 - Casing, Burster Assembly

PUBLICATIONS

Ordnance Corps
- ORD-M608-11 - Ordnance Inspection Handbook on Procedures and Tables for Continuous Sampling by Attributes

FSC 1315

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(Copies of Specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 General - Materials, parts and assemblies shall comply with the requirements specified on the applicable drawing, referenced specifications and the following:

3.2 Preproduction Sample - Unless otherwise directed by the contracting officer, a preproduction sample shall be required.

3.3 Mechanical Properties - Mechanical properties required by the applicable drawing shall be uniform throughout each component within specified tolerances.

3.4 Brazed Joint - The braze metal shall be visible for the full 360 degrees around the joint at juncture of casing and rear of sleeve.

3.4.1 Joint Section - When sectionalized, the braze metal joint shall be free of voids, cracks, entraped slag, cleaning or other fluids for the entire length of all visible areas of the joint section.

3.5 Workmanship - The requirements for workmanship are as shown on the applicable drawing, in the applicable specifications and the following:

3.5.1 Protective Coating - The protective coating shall be as specified on the applicable drawing and the specifications referenced thereon and shall apply to all indicated surfaces regardless of contour configuration.

3.5.2 Burr - No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.5.3 Foreign Matter - No part or assembly shall contain dirt, chips, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 General Quality Assurance Provisions - The quality assurance provisions in this section and other documents referenced in this specification are applicable to inspection performed by or for the Government. Reference shall be made to Standard MIL-STD-109 to define the inspection terms used. Inspection shall be in accordance with Specifications MIL-G-2550 and MIL-I-45208, except as specified herein.

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4.2 Preproduction Sample - Prior to initiation of regular production, the contractor shall submit a preproduction sample to a designated Government approved facility for evaluation. The sample shall consist of 25 burster assemblies as follows: 10 burster assemblies complete including paint, 5 burster assemblies complete less paint and 10 sleeves and casings unpainted. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials including packaging and packing must be the same as will be used for regular production, and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the preproduction sample shall be inspected for all requirements of the drawings and specifications. The Government reserves the right to require new preproduction samples until such time as an acceptable sample is submitted.

4.3 Inspection Provisions

4.3.1 Lot

4.3.1.1 Submittal of Product - The contractor shall submit the product in accordance with Standard MIL-STD-105 or ORD-M608-11, as applicable.

4.3.1.2 Size of Lots - For purposes of identification and shipment the lot size shall be as agreed upon between the contractor and the contracting officer, provided the homogeneity of the inspection lot has been retained.

4.3.1.3 Component Lots - Lots of casings or sleeves shall contain not more than one heat of steel.

4.3.2 Examination - One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105 or ORD-M608-11, as applicable. An AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect shall be used. All non-conforming material shall be rejected. Four power (4x) magnification may be used in performing visual examination for such defects as burr and foreign matter. The Ordnance Standard Finish Blocks prescribed by Standard MIL-STD-639 shall be used as a basis of comparison for surface roughness determination.

4.3.3 Classification of Defects

4.3.3.1 Sleeve, Dwawing C-10520018, covering a detail of drawing F-10520029.

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Defect	Method of Inspection
CRITICAL	
1. Metal Defective	Visual
MAJOR	
101. Flange not perpendicular with pressed fit area	Gage
102. Diameter, inside large	Gage
103. Diameter of undercut, min. (if applicable)	Gage
104. Length from rear of flange to bottom of small inside diameter	Gage
105. Length of undercut, max. (if applicable)	Gage
106. Surface finish, pressed fit area	Visual/Measure
107. Foreign matter large inside diameter	Visual
MINOR	
201. Burr	Visual
202. Foreign matter	Visual

4.3.3.2 Casing, Drawing C-10520017, covering a detail of Drawing F-10520029.

Defect	Method of Inspection
CRITICAL	
defined	
MAJOR	
101. Diameter, outside at front	Gage
102. Diameter, outside at rear	Gage
103. Thickness of wall at front for minimum length	Gage
104. Thickness of wall at rear of change in diameter	Gage
105. Thickness of base	Gage
106. Diameter change not within dimension	Gage
107. Metal defective	Visual
MINOR	
201. Length of area of change in diameters, max.	Gage
202. Burr	Visual
203. Foreign matter	Visual

4.3.3.3 Casing, Burster Assembly, Drawing D-10520029

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Defect	Method of Inspection
CRITICAL	
1. Longitudinal scratches on pressed fit area	Visual
2. Pressed fit area scored	Visual
3. Braze metal not visible for full 360 degrees at juncture of casing and rear of sleeve	Visual
4. Hydrostatic or air pressure stamp(s) missing	Visual
MAJOR	
101. Weight	Approved Scale
102. Concentricity of rear of casing with pressed fit area	Gage
103. Diameter of pressed fit area	Gage
104. Length from front face of flange to rear of pressed fit area	Gage
105. Length from rear of flange to rear of casing	Gage
106. Length total, max.	Gage
107. Reversed taper pressed fit area	Gage
108. Radius at rear of pressed fit area not properly blended	Gage
MINOR	
201. Diameter of mouth	Gage
202. Diameter of flange	Gage
203. Angle at rear of sleeve, incorrect	Gage
204. Radii or chamfer, missing or incorrect	Visual/Measure
205. Surface finish, improper (unless otherwise classified)	Visual/Measure
206. Protective coating, inadequate or damaged	Visual
207. Protective coating on sleeve beyond permissible length	Visual
208. Compound missing from pressed fit area	Visual
209. Stamping missing, incorrect or illegible	Visual
210. Burr	Visual
211. Foreign matter	Visual

4.3.3.4 Carton, Prior to sealing drawing.

CRITICAL

None defined

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Defect	Method of Inspection
MAJOR	
None defined	
MINOR	
201. Improper Assembly	Visual
202. Incorrect number of casings in carton	Visual
203. Separator missing or damaged	Visual

4.3.3.5 Packed and Sealed Carton, Drawing

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Carton damaged	Visual
202. Carton improperly sealed	Visual/Manual
203. Contents loose	Visual
204. Marking missing, incorrect or illegible	Visual

4.3.4 Inspection Testing - The following tests shall be performed in accordance with the provisions of 4.4, except as provided in Specification MIL-I-45208.

4.3.4.1 Mechanical Properties

4.3.4.1.1 Casing - One casing from each 5000 casings or fraction thereof in the lot shall be subjected to the tension test to determine compliance with the yield strength and elongation requirements of the applicable drawing. Failure of any specimen to comply shall be cause for rejection of the lot.

4.3.4.1.2 Sleeve - Two tension test specimens (from different bars) shall be selected from each heat of steel that is to be used in the manufacture of the sleeve and subjected to the tension test to determine compliance with the yield strength and elongation requirements of the applicable drawing. Failure of either specimen to comply shall be cause for rejection of the heat of steel represented by the sample.

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4.3.4.1.3 Strength of Assembly - After assembly and cleaning, but prior to gaging and the application of protective coating, each assembly shall be subjected to a hydrostatic test to determine compliance with the requirement of the assembly drawing. Failure to comply as evidenced by leakage, deformation or rupture shall be cause for rejection of the assembly. Assemblies that comply shall be stamped with the letter "H" as indicated on the assembly drawing, immediately after removal from the test fixture.

4.3.4.1.4 Air Test - Each burster casing that has passed the hydrostatic pressure test, after drying, shall be subjected to an air pressure test to determine compliance with the requirement of the assembly drawing. Failure of any assembly to comply as evidenced by a drop in gage pressure prior to minimum time limit shall be cause for rejection of the burster casing. Assemblies that comply shall be stamped with the letter "A" as indicated on the assembly drawing, immediately after removal from the test fixture.

4.3.4.1.5 Joint Section - Five burster casing assemblies from each lot after hydrostatic and air pressure testing but prior to application of protective coating shall be sectionalized to determine compliance with the requirements of the assembly drawing. Failure of any specimen to comply as evidenced by the presence of voids, cracks, entrapped slag, cleaning or other fluids in the brazing shall be cause for rejection of the lot.

4.3.4.2 Protective Coating - The sample size and acceptance criteria shall be in accordance with the assembly drawing and the requirements of the applicable specifications listed thereon. Failure to comply with the requirements of the applicable specification shall be cause for taking the corrective action recommended therein.

4.3.4.3 Straightness Inside - The inside diameter of each burster casing assembly, after painting shall be gaged as prescribed by the assembly drawing. Failure of the functional gage to enter to the full indicated length shall be cause for rejection of the assembly.

4.3.5 Inspection Equipment - The examination and tests shall be made using equipment approved by the Government.

4.4 Test Methods and Procedures

4.4.1 Mechanical Properties

4.4.1.1 Casing - The casings selected in accordance with 4.3.4.1.1 shall have the closed end removed. The cyldrincal portion shall be tested as a full length specimen. The plug used at small diameter end shall be inserted to a depth slightly beyond the change in dimension section. The procedures

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prescribed by Federal Standard No. 151 for determing yield strength and elongation shall be used. The limiting set shall be in accordance with the requirements of the applicable drawing.

4.4.1.2 Sleeve - From the samples selected in accordance with 4.3.4.1.2. Two type R-1 tension test specimens shall be taken. The procedures prescribed by Federal Standard No. 151 shall be utilized, the limiting set shall be in accordance with the requirements of the applicable drawing.

4.3.1.3 Strength of Assembly - The casing assembly shall be hydrostatically tested using equipment capable of performing the test accurately and consistently. The casing assembly shall be mounted in the test fixture and held in place by clamping the sleeve flange, no other portion of the assembly shall be restrained during the application of pressure. The pressure will be applied until minimum is reached, after which the pressure will be held for the minimum time indicated on the applicable drawing. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over 2 hours during operation, it shall also be calibrated prior to restarting the operation after a stoppage of 1 hour or more.

4.4.1.4 Air Pressure Test - This test shall be made using equipment capable of performing the test accurately and consistently. The casing assembly shall be mounted in an approved fixture equipped with an air pressure gage. The gage will be observed to determine that the proper pressure is applied, find indications of a drop in pressure before the minimum time limit prescribed by the assembly drawing. The equipment shall be calibrated at the start of each day's operation and at intervals of 4 hours, and after a stoppage of 3 hours or more.

4.4.1.5 Sectionalized Joint - The samples selected in accordance with 4.3.4.1.5 shall be sectionalized into 4 segments by cutting longitudinally along the assembly axis. Care shall be exercised to assure that no damage occurs to the brazed joint sections. The braze metal of the joints of each segment shall be visually examined to determine compliance with 3.5.1.

4.4.2 Protective Coating - The procedures for testing the cleaning of components and application of protective coating shall be in accordance with the criteria for these tests specified on the assembly drawing and in the specifications listed thereon.

4.4.3 Straightness Inside - The functional plug gage shall be entered into the mouth of the assembly, and must enter to the length indicated on the drawing.

5. PREPARATION FOR DELIVERY

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5.1 Packing

5.1.1 Level A - Not Applicable

5.1.2 Level B - Not Applicable

5.1.3 Level C - Burster casings shall be packed in accordance with MIL-P-10025.

5.2 Labeling and Marking - Labeling and marking for shipment shall be in accordance with MIL-P-10025.

5.3 Data Card - Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

NOTICE: When Government drawing, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility for any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any person or corporation or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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